

Scheme and Fantasy: Reading Usage, Not Box Scores

Fantasy points are the last link in a chain the play-caller forges: who is on the field, who runs a route, who the ball is aimed at, and how often the team throws at all

Week 1, Sunday night. You check your fantasy team. The tight end you drafted in the sixth round was on the field for 52 of his team's 61 snaps and caught one pass for 7 yards. Your league-mate's backup running back played 14 snaps and scored twice, both times from the 1-yard line. On Monday the talk is all touchdowns: the backup is a waiver-wire must, your tight end a bust. A coach looking at the same game would ask different questions. On how many of those 52 snaps did your tight end run a route, and on how many was he blocking a defensive end? Why was the backup the one in the game at the 1-yard line, and is that a role the play-caller will keep giving him? The box score records what happened. **Usage** (who is on the field, who goes out for a pass, who the ball is aimed at, how often the team throws at all) records what the scheme *intended* to happen, and intentions repeat from week to week far more reliably than outcomes. This chapter teaches you to read usage, to trace it back to the scheme and the play-caller who produce it, and to predict how it moves when the play-caller changes.

You'll be able to...

- Turn a scheme into receiver usage and read the numbers that measure it: route participation, target share, air-yards share, WOPR and yards per route run, and say which of them carry over from season to season.
- Read a running back's role (snap share, share of the carries, routes run, red-zone share) and a tight end's role in 12 personnel (route runner or extra blocker), and spot the goal-line role that makes or breaks a back's touchdown total.
- Forecast a player's volume from his team's tendencies: pass rate over expected, pace and game script.
- Explain the common scoring formats (standard, half-PPR, PPR), what each one rewards, and why average draft position works like a market price.
- Predict how a coordinator change shifts fantasy value, using the 2026 changes in Detroit, Los Angeles (Chargers) and Kansas City as test cases.

You'll need

This is the second chapter of Part 14, and it leans on Parts 2, 4 and 13. One line each, with a link if you want to catch up:

- **Building a roster** (Building a Roster): positional value (how much a position, as opposed to a player, contributes to winning; running backs and tight ends sit near the bottom of the real-football market), surplus value (what a player produces minus what he costs) and scheme fit (the same player is worth more in one system than another). Fantasy has its own market, and this chapter borrows all three ideas for it.
- **Tendencies from data** (Tendencies: PROE, Personnel, Motion, and Coverage): xpass is a model's estimate of how likely an average team is to pass in a given situation (down, distance, field position, score, time), and PROE (pass rate over expected) is how much more or less often a team actually throws than that, usually measured in neutral situations (early downs, game still in doubt, not the last two minutes of a half) so the score doesn't drive it.
- **The data** (Working with nflverse Data in Python): play-by-play records every play; participation data lists the 22 players on the field for each play (2016-2025; the current season arrives only after it ends); and FTN charting data flags motion, play-action and the quarterback's alignment, play by play, updated weekly.
- **Personnel and the passing game** (Personnel Groupings, Passing Fundamentals): 12 personnel is one back, two tight ends and two receivers; air yards are how far past the line of scrimmage a pass travels in the air, and aDOT is a receiver's average.

Reading the diagrams: offense is blue circles (X and Z wide receivers; Y and U tight ends, ringed in gold; F a fullback; Q quarterback; R running back); defense is outlined squares. Solid arrows are routes and runs, short arrows ending in a bar are blocks, dashed lines defenders' movements. The dark blue line is the line of scrimmage. The charts come from nflverse data; the notes under each one say which seasons and filters it uses.

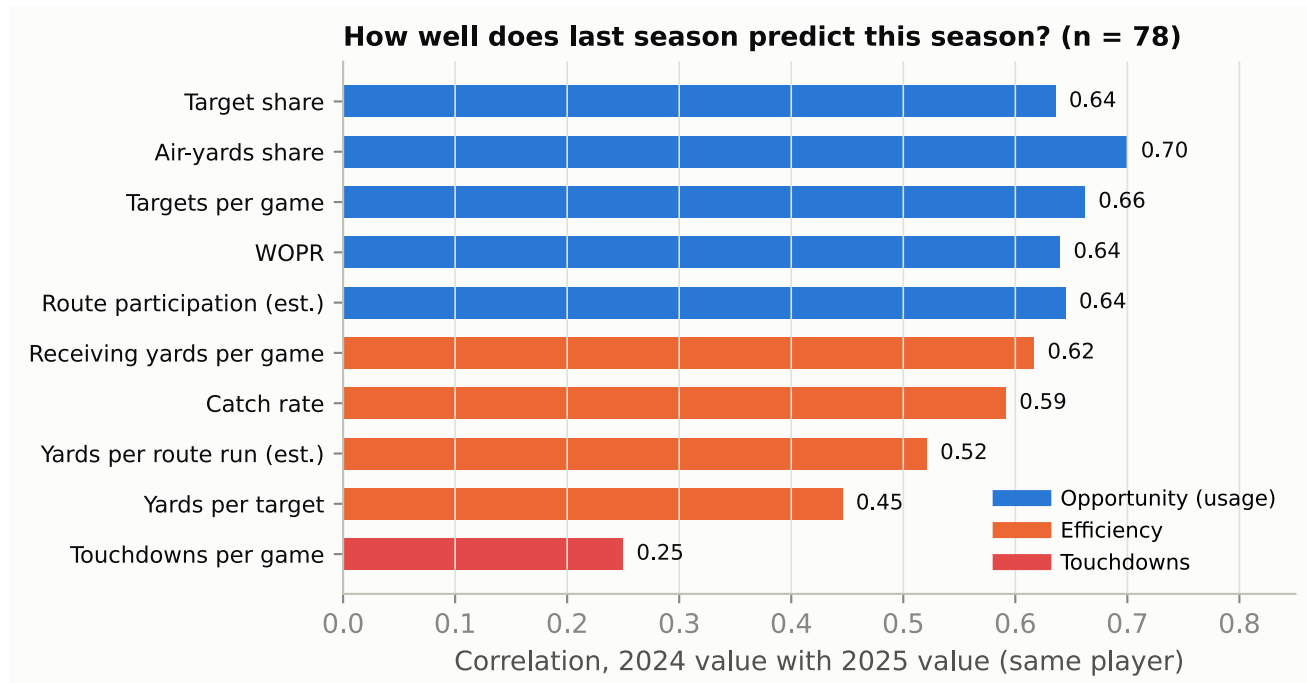


Figure 1: Usage repeats; touchdowns don't. Each bar is the correlation between a player's 2024 and 2025 value of the statistic, for the 78 wide receivers and tight ends with 50 or more targets in both regular seasons (a player who changed teams counts with the team where he drew more targets). A correlation of 1 would mean the 2024 number predicts 2025 perfectly; 0, not at all. Blue bars measure opportunity, orange bars what the player did with it, the red bar touchdowns. Route participation and yards per route run are estimated from on-field dropbacks (see the note on where route counts come from). Data: nflverse play-by-play and participation, this book's calculation.

Outputs and inputs: why the box score misleads

A fantasy point is the end of a chain. For a wide receiver, the chain runs like this: his team runs a certain number of plays (**pace**); some fraction of them are passes (the play-caller's tendency, adjusted by the score); on some of those passes he is on the field and running a route rather than blocking or standing on the sideline (**route participation**); on some of those routes the ball is thrown to him (**target share**), some of them deep (**air-yards share**); and on some of those targets he catches the ball, gains yards and, now and then, scores. Every link before the catch is mostly **usage**, and usage is set by people: the coordinator who designs the plays, the play-caller who chooses them on game day, and the quarterback who decides where the ball goes. The links after it are mostly **efficiency**, which belongs to the player, plus a large dose of luck.

The reason to care about the difference is that the two halves behave very differently over time. The figure below takes every wide receiver and tight end with at least 50 targets in both 2024 and 2025 and asks a simple question of each statistic: if you knew a player's 2024 number, how well would it predict his 2025 number?

The opportunity numbers (target share, air-yards share, targets per game, WOPR) all correlate at about 0.65 to 0.7 from one season to the next. The efficiency numbers sit lower: yards per target is down near 0.45. Touchdowns per game, the statistic that swings a fantasy week more than any other, correlate at 0.25. In plain terms, a receiver's share of his team's targets is mostly a fact about his role, and his role is decided before the season by the scheme, the depth chart and the quarterback's trust. Whether a given target turns into a 40-yard catch or a touchdown depends on the coverage, the throw, the defender's hand and a dozen things that don't repeat.¹

¹Course calculation from nflverse regular-season play-by-play (2024-2025) and `load_participation()` (FTN-sourced for both seasons): wide receivers and tight ends with 50 or more targets in both seasons (n = 78); for a player with two teams in a season, the team where he drew more targets. Target share and air-yards share are measured against the team's totals in the games the player was on the field; two-point tries excluded. "Route participation (est.)" is the share of the team's dropbacks (`qb_dropback == 1`, which includes sacks and scrambles) on which the player was listed on the field, and "yards per route run (est.)" divides receiving yards by those dropbacks; true route counts come from PFF or FTN film charting and are lower for anyone who sometimes blocks.

! Why it exists

Usage stats exist because fantasy players and analysts kept being fooled by box scores. A season of touchdowns makes a player look like a star; a season of near-misses makes an equally used player look ordinary. The next season, the touchdowns are dealt out again more or less at random, while the opportunity that produced them carries over. Reading usage is the habit of pricing a player on the part of his production that will repeat. The trade-off: usage tells you about the chances, not the talent. A player who gets the ball constantly and does nothing with it will eventually lose the role, so the efficiency numbers still matter. They are just slower to trust.

i Madden vs. real life

In Madden, a player's ratings drive what he does on the field: a 95-overall receiver gets open and makes plays. In a real offense, the ball goes where the play was designed to send it, read by the quarterback in the order the play-caller drew it. A great receiver in a run-first offense that rotates its receivers can finish behind an average one in a pass-first offense that never takes him off the field. The talent is the same; the opportunity is not, and opportunity is drawn on the play-caller's call sheet, not in the player's ratings.

Receivers: from snaps to routes to targets

Being on the field is not the same as running a route

Start with the first filter. A player can't catch a pass on a play he isn't in for, so the first number to check is **snap share**: the share of the team's offensive plays on which he is on the field, published by the league in its official snap counts. A team's top two wide receivers usually play 85-95% of the snaps; a third receiver might play 60-80%, and his number depends on how often the team uses three-receiver sets.

Snap share has a blind spot, and the tight end shows it best. The figure below draws the same 12 personnel formation twice.

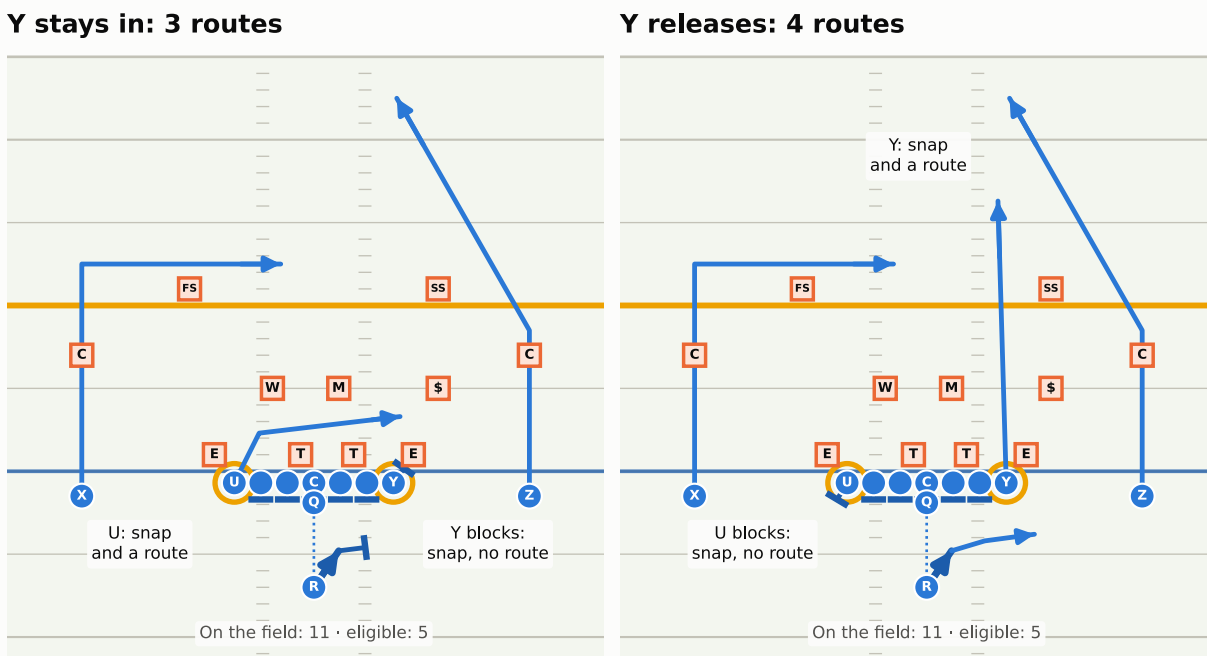


Figure 2: One snap, two jobs (illustrative play-action pass from 12 personnel, under center). Both tight ends (gold rings) are on the field for both plays, so both get a snap in the official count. Left: Y stays in to block the defensive end and the running back fakes the run and then protects, so only three players run routes: X, Z and U. Right: the same formation, but U stays in and Y runs up the seam while the back releases to the flat after checking for a blitz, so four players run routes. Snap share can't tell these plays apart; route participation can.

In both pictures the two tight ends play the snap, so their snap counts are identical. In one of them Y is a blocker who never had a chance to catch the ball; in the other he is a receiver running up the seam. Over a season, a tight end who stays in to block on a third of his team's dropbacks has a third fewer chances to be targeted than one who goes out every time. The same is true of a running back who stays in to protect the quarterback, a job called pass protection, or who chips a pass rusher before releasing late.

That is what **route participation** measures: the share of the team's dropbacks (pass plays, including sacks and scrambles) on which a player runs a route. For a team's top wide receivers it is usually above 85%, because a wide receiver almost never stays in to block on a pass play. For tight ends and backs it is where the real information is: two tight ends with the same snap share can have route participation of 50% and 85%, and only the second is a fantasy tight end.

i Go deeper: where route counts come from

Routes are not in the official play-by-play. Pro Football Focus (PFF) and FTN count them by watching film of every play, and sell the results. The free nflverse participation data tells you who was on the field for each play, but not who ran a route. This chapter therefore *estimates* route participation as the share of a team's dropbacks on which a player was on the field. For wide receivers the estimate is close to the true figure; for tight ends and backs it overstates routes by however often they stayed in to block. When this chapter says "route participation (est.)", that is the measure. Treat it as an upper bound for anyone who blocks.

Target share: how much of the passing game is aimed at him

A **target** is a pass thrown to an intended receiver, caught or not. **Target share** is a player's targets divided by his team's targets in the games he played. It is the single most useful receiving number, for two reasons. First, it is a share, so it strips out how much the team throws: 25% of a pass-heavy team's targets and 25% of a run-heavy team's targets describe the same role in two different amounts of work. Second, it is set mostly by design. A play-caller who builds his passing game around one receiver gives that receiver the first read on many concepts, and the quarterback, reading the play in order, throws to him whenever he is open.

In 2025 a typical team's leading receiver drew about 23% of its targets in the games he played; the true "alpha" receivers were near or above 30%. Seattle's Jaxon Smith-Njigba, the 2025 Offensive Player of the Year, drew 36%, the highest share of any player with a full season, and Detroit's Amon-Ra St. Brown and Cincinnati's Ja'Marr Chase each drew about 31-32%. A player in the 15-20% range is a secondary option; his weekly output depends on how the defense plays the primary one.

Air-yards share and WOPR: not all targets are equal

A 2-yard pass in the flat and a 30-yard throw down the sideline each count as one target, but they are not worth the same. The deep one, caught, is worth far more yards and points. **Air-yards share** captures that: add up the air yards of every pass thrown to the player (how far past the line of scrimmage each traveled in the air) and divide by the team's total. A slot receiver who catches short passes can have a big target share and a modest air-yards share; a deep threat can have the reverse.

Josh Hermsmeyer, an analyst who built the air-yards site airyards.com and later wrote for FiveThirtyEight, combined the two into one number: **WOPR**, the weighted opportunity rating.²

$$\text{WOPR} = 1.5 \times \text{target share} + 0.7 \times \text{air-yards share}$$

The weights come from fitting the two shares to receivers' fantasy production; target share gets the bigger weight because a target is a chance at a reception, yards and a touchdown all at once, while air yards add the size of the chance. As a scale, a WOPR of about 0.7 or more marks a team's clear number-one target; 0.4 to 0.5 is a typical number two. In 2025 Smith-Njigba's was about 0.89 (a 36% target share and half of Seattle's air yards), the highest in the league. Carolina's Tetairoa McMillan, the Offensive Rookie of the Year, drew a quarter of the Panthers' targets and about 43% of their air yards for a WOPR near 0.68, among the league's top ten: a rookie who was the deep, contested-catch target from his first game.

Yards per route run: efficiency with the right denominator

Opportunity is half the story; the other half is what a player does with it. The classic efficiency numbers (catch rate, yards per catch, yards per target) all divide by something the player earned only after he got open. **Yards per route run (YPRR)** divides

²The formula as implemented in nflverse: "wopr: Weighted Opportunity Rating, $1.5 \times \text{target_share} + 0.7 \times \text{air_yards_share}$," nflreadr data dictionary, player stats: https://nflreadr.nflverse.com/articles/dictionary_player_stats.html. Hermsmeyer as its creator, alongside RACR and PACR: Action Network, "WOPR" explainer: <https://www.actionnetwork.com/article/p/601206703>; Dynasty Nerds, <https://www.dynastynerd.com/?p=33320>; 4for4 author page ("creator of airyards.com and Air Yards based metrics PACR, RACR and WOPR"): <https://www.4for4.com/users/josh-hermsmeyer/author-page>. Hermsmeyer's original airyards.com write-up is no longer online; the weights here are the ones every secondary source and nflverse use.

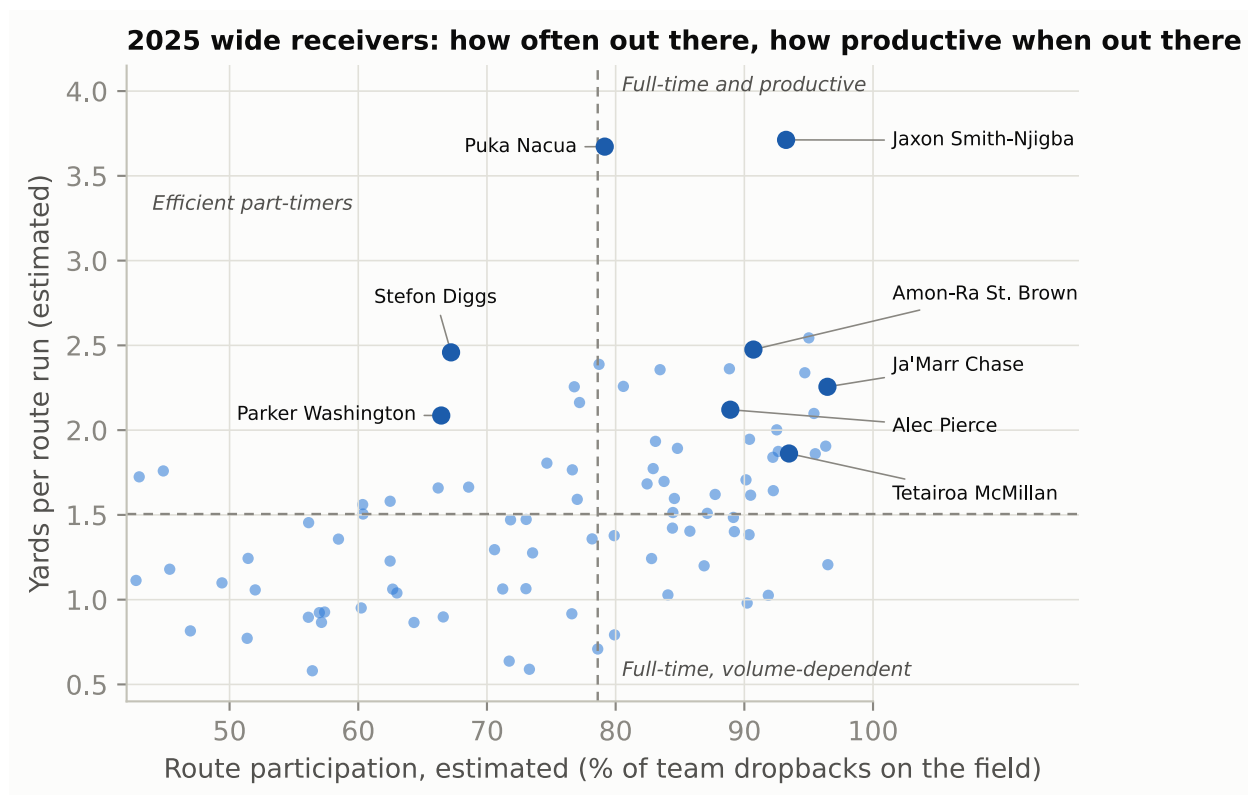


Figure 3: Route participation against yards per route run, 2025 wide receivers with 250 or more dropbacks on the field (estimated from participation data: the share of the team’s dropbacks, in games he played, on which he was on the field; and receiving yards divided by those dropbacks). Dashed lines are the medians (about 79% and 1.5). Top right: full-time players who produce on every route, the safest fantasy receivers. Top left: efficient part-timers, whose value rises if their role grows. Bottom right: full-time players whose production per route is modest; they depend on volume. Regular seasons only.

receiving yards by the number of routes the player ran, every chance he had to get open, whether or not the ball came his way. It rewards both halves: a receiver who earns targets *and* gains yards on them scores well, and one who runs a hundred routes for nothing scores badly. Around 1.5 is typical for a regular NFL wide receiver; 2.0 is very good; 2.5 or more is elite.

The two numbers together, how often a receiver is out there and how productive he is when he is, form a simple map.

Smith-Njigba and the Rams’ Puka Nacua sit alone at the top, at about 3.7 yards per route, numbers that happen a few times a decade. The more interesting dots for fantasy are the ones in the top left: New England’s Stefon Diggs produced almost 2.5 yards per route while on the field for only about two-thirds of the Patriots’ dropbacks. A part-timer that productive is the cheapest kind of upside: if an injury or a new coordinator turns him into a full-timer, his volume rises by a third and his efficiency doesn’t have to improve at all. The bottom right holds the opposite profile, full-time receivers who are useful only as long as the volume keeps coming.

🔥 Common misconception

“He had 9 catches for 140 yards, so he’s the number-one receiver now.” One big game is mostly efficiency, and efficiency is the noisy half. Check the usage behind it: was he on the field for more routes than usual, and did his target share actually rise? If he ran his usual 60% of routes and three long passes happened to find him, nothing about his role changed. If his route participation jumped from 60% to 90% because the starter went down, the role changed, and that is worth acting on even if the box score was quiet.

Running backs: snaps, carries, routes and the goal line

A running back’s usage has more parts than a receiver’s, because he does more jobs. Read it in four pieces.

1. **Snap share.** Very few backs play every down now. A true “bell cow” plays 75-85% of the snaps; most backfields split, with an early-down back and a passing-down back sharing 50-60% each.

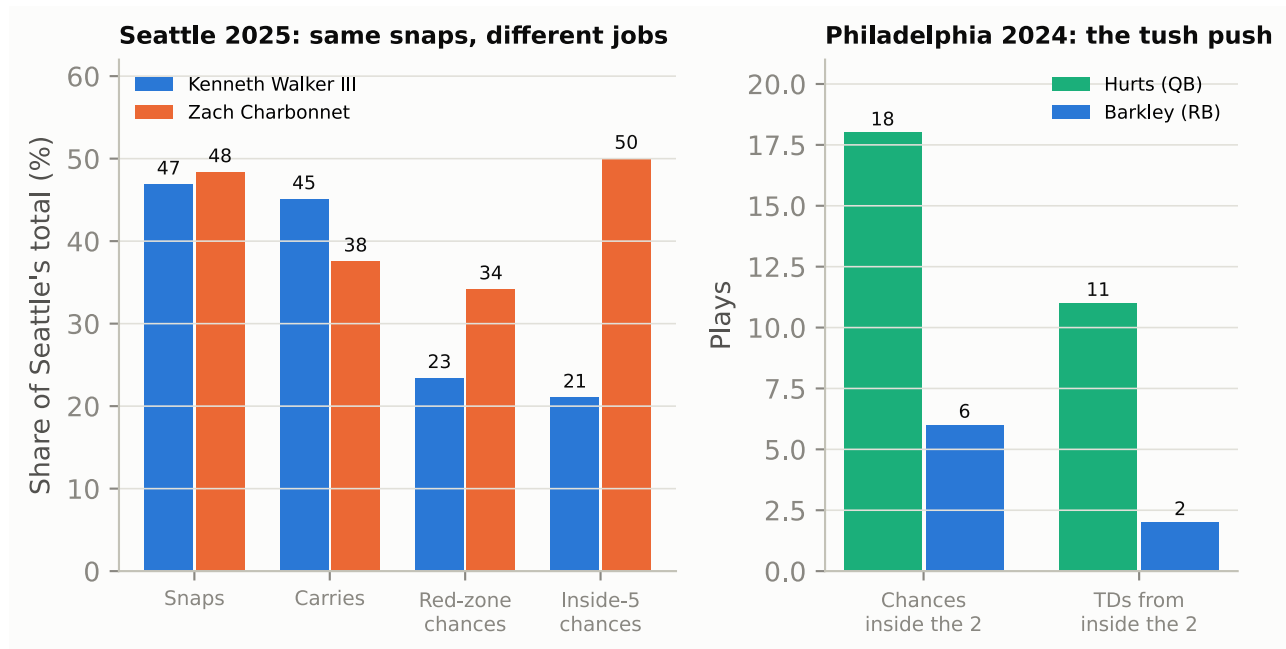


Figure 4: Who gets the ball near the goal line. Left: Seattle’s 2025 backfield. Kenneth Walker III and Zach Charbonnet were on the field for about the same share of snaps, and Walker got more of the carries, but Charbonnet got about half of the team’s chances inside the 5-yard line and scored 12 rushing touchdowns to Walker’s 5. Right: Philadelphia, 2024. Saquon Barkley ran for 2,005 yards, but inside the 2-yard line the Eagles gave the ball to quarterback Jalen Hurts on the tush push three times as often, and Hurts scored 11 times from there to Barkley’s 2. ‘Chances’ are carries plus targets; snap shares are game averages from the official counts. Data: nflverse play-by-play and snap counts (Pro Football Reference), regular seasons.

2. **Share of the carries.** How many of the team’s runs he gets. This depends on the team’s run rate as well as his role, so check both.
3. **Routes and targets.** A back who runs routes on third down and in the two-minute drill gets targets that a pure runner never sees, and in PPR scoring (below) each catch is worth a point. Backs run routes on far fewer dropbacks than receivers, because they often stay in to protect.
4. **Red-zone share.** His share of the team’s carries and targets inside the opponent’s 20-yard line, and above all inside the 5. Most touchdowns are scored from close range, so the goal-line role decides a back’s touchdown total more than his talent does.

The last one is where scheme hides the most fantasy value, and two recent backfields show it from both directions.

Seattle’s split is the classic two-back design. Walker was the early-down runner who got more of the carries between the 20s; Charbonnet, a capable short-yardage runner, was the back the coaches trusted near the goal line. Walker’s 221 carries brought 5 rushing touchdowns; Charbonnet’s 184 brought 12, and in standard scoring Charbonnet out-scored him for the season. The snap share said “even split.” The red-zone share said who would score.

i Film room: Eagles 2024 — Saquon Barkley's 2,005 yards

Philadelphia signed Saquon Barkley as a free agent from the Giants in March 2024. Fantasy drafters were wary: he was 27, coming off a down year behind a poor line, and joining a team whose quarterback scored short touchdowns himself. In July his consensus half-PPR average draft position was about the 17th pick, the lowest of his career.³

What he joined was a scheme built to feed a back. The Eagles' line was the best-funded in football (see Building a Roster and the Eagles case study), and the play-callers leaned on it: in neutral situations the 2024 Eagles passed about 9 percentage points less often than an average team in the same situations, the third-lowest PROE in the league. They also led often, and a team protecting a lead runs. Barkley took 58% of the Eagles' runs and played about three-quarters of the snaps, and he finished with 2,005 rushing yards on 345 carries and 13 rushing touchdowns in 16 games, sitting out Week 18 with the Eagles already locked into the NFC's No. 2 seed.⁴

Now look at what he *didn't* get. His 43 targets were modest for a lead back, because Philadelphia's passing game ran through A.J. Brown and DeVonta Smith. And at the goal line the ball went to the quarterback: the Eagles' **tush push**, a quarterback sneak with teammates pushing from behind (built in full in Red Zone, Goal Line, and Short Yardage), was so reliable that inside the 2-yard line Hurts got 18 chances to Barkley's 6 and scored 11 times to Barkley's 2. A back with 2,005 yards finished the season first among running backs in standard scoring but second in PPR, behind Detroit's Jahmyr Gibbs, who ran less but caught more.

What to notice: every input that made the season was visible before it started: a run-first play-caller, an elite line, a team expected to lead. So was the one that capped it, the quarterback's goal-line role. The market priced the worry (the goal line) and underpriced the volume.

The tight end in 12 personnel: blocker or receiver?

No position's fantasy value depends more on scheme than the tight end's, because no position's job varies more. In 12 personnel (one back, two tight ends, two receivers) the two tight ends usually have different jobs. One is the in-line tight end, attached to the end of the line, whose first job is blocking; the other is often the move tight end, who lines up in the slot, as a wing or in the backfield and is a receiver first. Offenses used 12 personnel on 24.4% of plays in 2025, a decade high in this book's count, and fewer than three wide receivers on 41.7% of plays by Next Gen Stats' count.⁵ More tight end snaps do not mean more tight end routes: the second tight end in a heavy set is often there to block, which is exactly why the offense put him there instead of a third receiver.

So for a tight end, read three numbers in order.

- **Route participation, not snap share.** A tight end who runs a route on 80% or more of dropbacks is used as a receiver; one at 50-60% is a blocker who sometimes goes out. Fantasy points live almost entirely in the first group.
- **Target share among route runners.** Some offenses aim the passing game at the tight end. In 2025 Arizona threw 35% of its targets to tight ends, the most in the league, and Trey McBride alone drew 27% of the Cardinals' targets, more than any wide receiver on the team.
- **Red-zone usage.** Tight ends are big targets in tight spaces, and some offenses build their goal-line pass plays around them. A tight end with few yards but a steady red-zone role is a touchdown bet, with the noise that implies.

³FanDuel Research, "Fantasy Football: It's Never Been Easier to Draft Saquon Barkley" (July 2024): <https://www.fanduel.com/research/fantasy-football-it-s-never-been-easier-to-draft-saquon-barkley> (FantasyPros consensus half-PPR ADP, pick 17, "his lowest ADP in seven seasons," i.e. of his career; the article uses July 2024 ADP data). ADP kept moving in August, so treat 17 as the July price.

⁴Course calculation from nflverse 2024 regular-season play-by-play and `load_player_stats()`: 345 carries, 2,005 yards, 13 rushing touchdowns in 16 games; 33 catches on 43 targets for 278 yards and 2 touchdowns; 58% of the Eagles' runs (scrambles included); 322.3 standard and 355.3 PPR fantasy points (first and second among running backs). Eagles' neutral PROE -8.8, third-lowest. Snap share: Pro Football Reference snap counts via nflverse, 73% average. Inside the 2-yard line: Hurts 18 carries or targets and 11 touchdowns, Barkley 6 and 2. Pro Football Reference: <https://www.pro-football-reference.com/players/B/BarkSa00.htm>. Week 18: Nick Sirianni announced that Barkley and other starters would sit against the Giants with Philadelphia locked into the No. 2 seed (Detroit and Minnesota were playing for No. 1), leaving him 100 yards short of Eric Dickerson's record: NBC Sports Philadelphia, <https://www.nbcsportsphiladelphia.com/nfl/philadelphia-eagles/eagles-inactives-saquon-barkley-officially-inactive-nfl-week-18/640273>; SI, <https://www.si.com/nfl/eagles/onsi/news/eagles-saquon-barkley-gets-honest-about-sitting-week-18-pat3>.

⁵12 personnel share: course calculation from nflverse participation data, regular-season pass and run plays (see Personnel Groupings). Fewer than three wide receivers: NFL Next Gen Stats via NFL.com, "NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks": <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

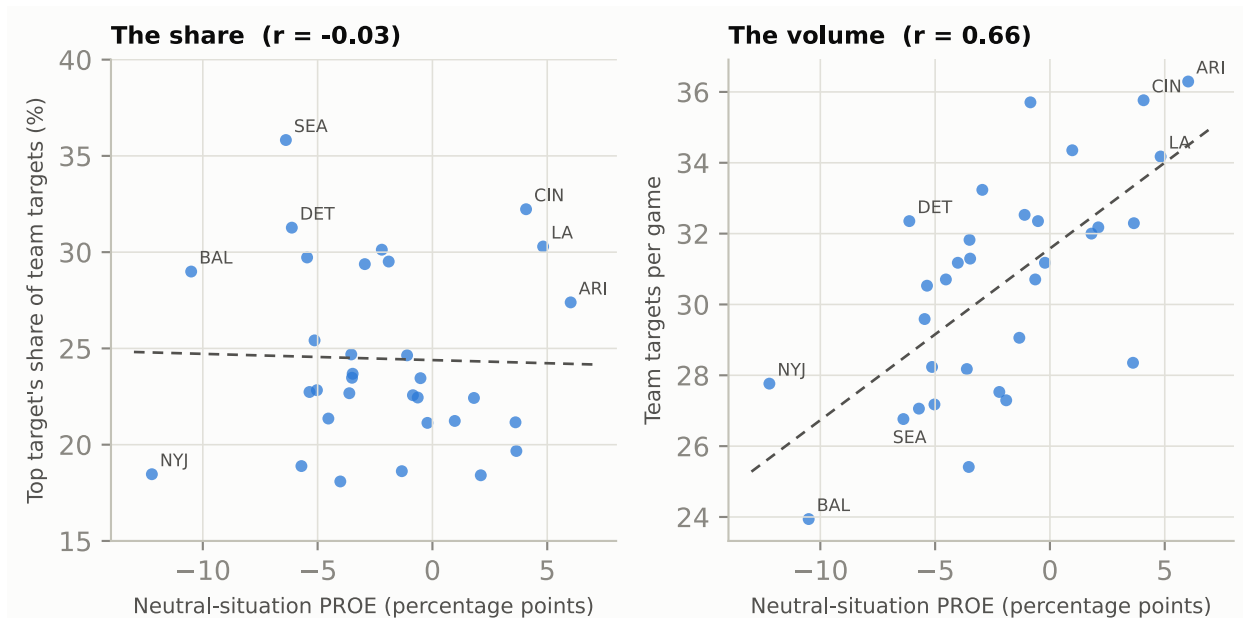


Figure 5: PROE moves the volume, not the share. Each dot is one team in 2025. Left: the share of the team's targets that went to its most-targeted player, against the team's neutral-situation PROE. There is no relationship (correlation about 0): how pass-heavy a team is says nothing about how concentrated its targets are. Right: the team's targets per game against the same PROE. Here the relationship is strong (correlation about 0.66): pass-heavy teams throw more passes, so a given target share is worth more targets. PROE is the average of nflverse's pass_oe on downs 1-3 with win probability between 20% and 80%, outside the last two minutes of each half. Regular season.

The Arizona number matters for 2026, because the coordinator who built that offense now calls plays in Detroit. We'll come back to him.

Volume: how much the team throws, how fast, and why the score changes it

Shares tell you a player's slice. **Volume** is the size of the pie, and it is set at the team level by three things: how pass-happy the play-caller is, how many plays the offense runs, and the score.

Pass rate over expected

PROE measures the first. It compares how often a team passes to how often an average team would pass in the same down, distance, field position, score and time, so it isolates the play-caller's taste from the situations he was handed. The figure below shows why it matters for fantasy, and what it does *not* do.

The left panel is the reassuring one: a receiver's slice doesn't depend on whether his team likes to throw. Seattle (PROE about -6) and the Rams (about +5) both funneled their passing games through one receiver, while the Jets, the most run-heavy team in neutral situations, spread theirs thin. The right panel is the one that moves fantasy values. Arizona, the league's most pass-happy team at about +6, threw about 36 targets a game; Baltimore, at about -10, threw about 24. The same 25% target share is 9 targets a game in Arizona and 6 in Baltimore, a 50% difference produced entirely by the play-caller's taste.

That gives the basic volume equation for any receiver:

$$\text{targets per game} \approx \text{plays per game} \times \text{pass rate} \times \text{target rate} \times \text{target share}$$

where the pass rate is set by PROE plus the situations the team gets into, and the "target rate" (the share of pass plays that produce a target, rather than a sack, scramble or throwaway) is usually around 85-90%. A coordinator change can move every term except the last one at a stroke.

Pace

Pace is the number of plays. In 2025 the slowest offenses (Miami, Minnesota and Las Vegas) ran about 55 plays a game; the fastest (Dallas and Chicago) ran about 64-65. Ten extra plays a game is roughly 15-18% more opportunity for every player on the offense, before anyone's share changes. Pace has two sources: how quickly a team gets to the line and snaps the ball (its tempo, built in In-

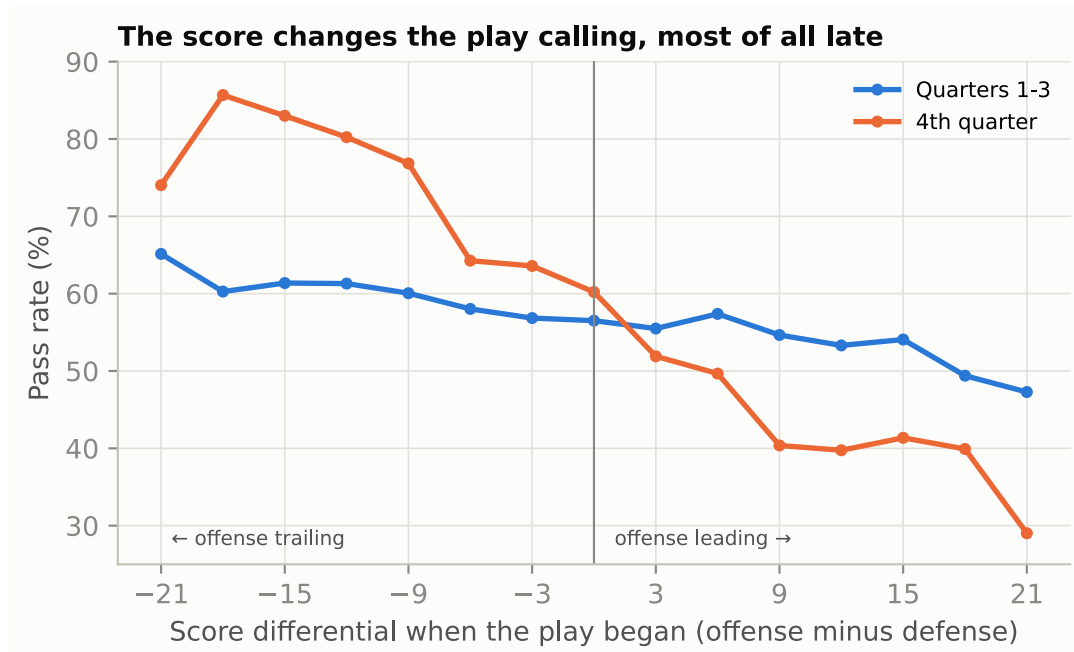


Figure 6: Game script: how often teams passed at each score, 2025. Each point is the pass rate (dropbacks, including sacks and scrambles, as a share of plays) for plays with that score differential from the offense’s point of view, grouped in 3-point steps and capped at 21 either way; the last two minutes of each half are excluded because the clock dominates them. In the first three quarters the line is fairly flat, from about 65% when far behind to about 47% when far ahead. In the fourth quarter it tilts sharply: about 80% when behind by two scores, about 40% or less when ahead by two scores. Regular season.

Game Adjustments and Tempo), and how many first downs it earns, since drives that keep moving produce more plays. A good defense helps too: an offense gets more possessions when its own defense gets off the field fast.

Game script

The score is the last term, and it is the one fantasy players talk about most. **Game script** is the way the score shapes play calling. A team ahead late runs the ball, which keeps the clock moving (an incomplete pass stops it), protects against turnovers and wears down a defense; a team behind throws, because it needs points fast and passes gain more per play. The effect is large enough to swamp a play-caller’s taste.

For fantasy, game script is why a receiver on a bad team can be a good pick (his team will trail, and trailing teams throw) and why a running back on a good team can be one (his team will lead, and leading teams run). It is also why one game’s usage can mislead: a running back who got 28 carries in a 31-3 win got them because of the score, not because his role grew, and the week his team trails he may get 11. When you read a weekly usage line, ask what the score was. The cleanest usage numbers come from the stretches when the game was close, the same reason PROE is measured in neutral situations, and why analysts drop garbage time when they measure a team.

Common misconception

“Pass-heavy teams are always better for receivers.” Over a season, mostly yes, because PROE drives volume. But a team’s *total* passing volume also rises when it is bad and trailing, and those extra targets come late, against defenses playing soft coverage to protect a lead. A receiver who racks up yards when his team is down three scores is real for fantasy (the points count), but the same team in a close game might throw much less. When you project volume, look at the team’s PROE (its taste) and its likely score (its fate) separately.

Scoring formats and the market

Standard, half-PPR and PPR

Fantasy football began as a touchdown game. The first league, the Greater Oakland Professional Pigskin Prognosticators League, was dreamed up in 1962 by Bill Winkenbach, a limited partner in the Oakland Raiders, and held its first draft in 1963, with scoring

based on touchdowns and kicks.⁶ As statistics became easy to collect, leagues added yards, and the scoring that most hosting sites called “standard” settled into one point per 10 rushing or receiving yards, six per touchdown, one per 25 passing yards and four per passing touchdown, minus two for an interception or a lost fumble.

Standard scoring had a problem: it made running backs who scored touchdowns far more valuable than anyone else, and it made a 9-catch, 70-yard day by a slot receiver worth less than a single 1-yard touchdown run. **PPR** (point per reception) scoring adds one point for every catch, and half-PPR adds half a point. It spread through serious leagues in the 2000s, and a version of it is now the default at the major hosting sites (full PPR at ESPN, half-PPR at Yahoo).⁷ The change is not cosmetic: it moves the whole market toward the players whose usage is receptions.

Read the format before you read the usage. In standard scoring, a back’s carries and goal-line role are nearly everything; in PPR, a back who runs routes on third down can out-score a back with twice his carries. In 2025 Pittsburgh’s Kenny Gainwell was the 24th-best running back in standard scoring and the 16th in PPR, a jump bought entirely with 73 catches. Indianapolis’s Jonathan Taylor went the other way: first in standard, fourth in PPR, because 18 rushing touchdowns count the same in both systems while his 46 catches counted for less than Christian McCaffrey’s 102. McCaffrey, the Comeback Player of the Year, was the only back who did both, and he finished first in PPR.

ADP: the market price

ADP (average draft position) is the average pick at which a player is taken across many drafts on a hosting site. It is the closest thing fantasy has to a market price, and it behaves like one: it aggregates thousands of opinions, reacts to news and overreacts to last season’s box scores.

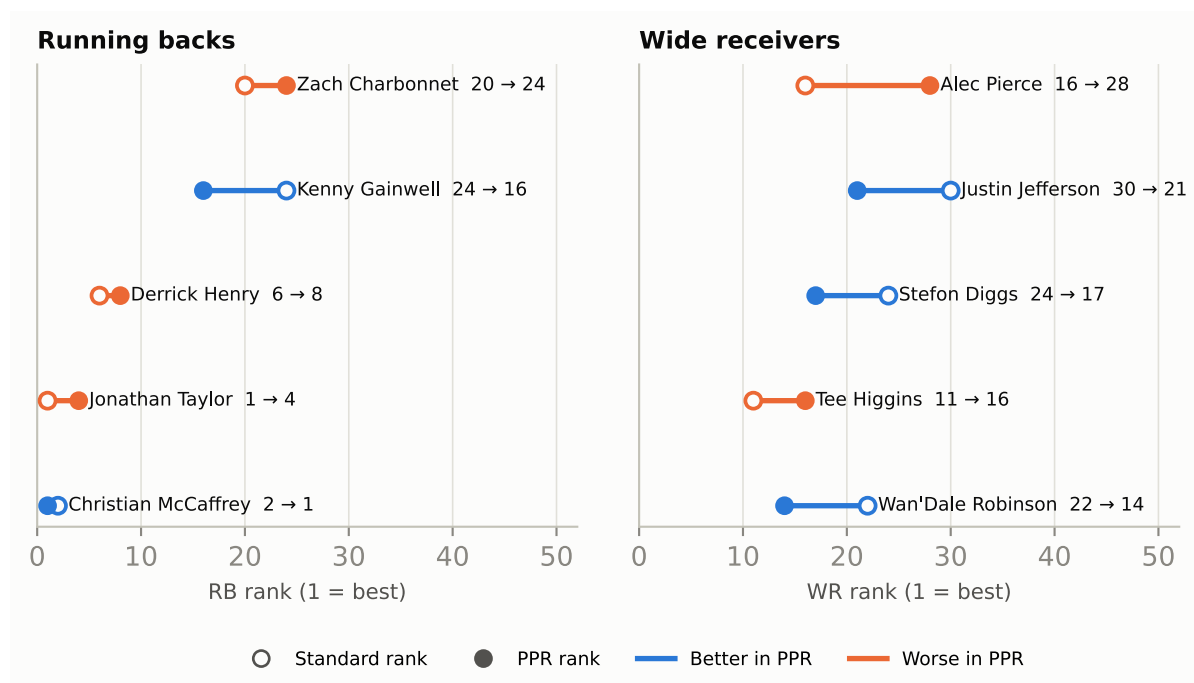


Figure 7: The same 2025 season under two scoring systems. Each line runs from a player’s position rank in standard scoring (open circle) to his rank in PPR (filled circle); rank 1 is the best at his position. Players who catch a lot of short passes (Kenny Gainwell, a pass-catching back with 73 catches on only 114 carries; Wan’Dale Robinson, a 140-target slot receiver) climb in PPR. Touchdown-heavy or deep-ball players (Jonathan Taylor, 18 rushing touchdowns; Alec Pierce, a deep threat with 84 targets and 47 catches) fall. Fantasy points as computed by nflverse (load_player_stats), regular season.

⁶“Fantasy football (gridiron),” Wikipedia: [https://en.wikipedia.org/wiki/Fantasy_football_\(gridiron\)](https://en.wikipedia.org/wiki/Fantasy_football_(gridiron)) , on Wilfred “Bill” Winkenbach and the GOPPPL, whose rules were worked out in 1962 at a New York hotel during a Raiders road trip, with a first draft in August 1963; NFL.com, “Let’s play GOPPL: Digging out the roots of fantasy football”: <https://www.nfl.com/news/let-s-play-goppl-digging-out-the-roots-of-fantasy-football-09000d5d80874333> . The original scoring paid only for touchdowns (rushing, receiving or passing), field goals, extra points and return touchdowns; yardage came later.

⁷RotoWire, “What is PPR in fantasy football?”: <https://www.rotowire.com/football/article/what-is-ppr-in-fantasy-football-94969> ; Sports Illustrated, “Fantasy Football 101: What Are PPR (Point Per Reception) Leagues?”: <https://www.si.com/fantasy/fantasy-football-101-what-are-ppr-point-per-reception-leagues-01k0y4hz2k7n> . The 2000s spread: the high-stakes NFLC adopted PPR at its founding in 2004 (founder Greg Ambrosius, NFLC forums: <https://nflcforums.shgn.com/viewtopic.php?p=46813>), and Footballguys added PPR scoring and rankings in 2008 (https://www.footballguys.com/article/Pasquino_Points_Per_Deception_2014). Site defaults: “ESPN’s default is full-point PPR, Yahoo’s is .5 PPR,” RotoWire: <https://www.rotowire.com/football/article/weekly-rankings-41764> .

The useful way to think about it borrows the idea of **surplus value** from Building a Roster. An NFL team doesn't profit from drafting a good player; it profits from drafting one who outplays the slot he cost. A fantasy pick works the same way. The 5th running back taken who finishes 5th was fairly priced. Barkley in 2024, taken around the 17th pick and finishing first in standard scoring, was a bargain. The way to find bargains is to find usage the market hasn't priced: a role that changed in a way the box score hasn't shown yet, a team whose volume is about to rise, a goal-line job that just opened. And the most predictable way roles change is a new play-caller.

! Why it exists

Every fantasy league is a small market with a fixed supply of players, and ADP is how the market's beliefs become visible. It exists because drafts on the big hosting sites are public data, and averaging them reveals the consensus. The trade-off for you as a drafter: following ADP keeps you from big mistakes, but it also guarantees you no edge, because you're paying the consensus price. Edge comes from disagreeing with the market for a reason the market is slow to see, and scheme is the slowest thing for a box-score market to see.

When the coordinator changes

An offensive coordinator is not a neutral pipe between the players and the box score. Each one has a fingerprint: how often he throws, how fast he plays, how much he uses motion and play-action, how much he aims at the tight end or the backs. Those habits travel with him from team to team, more faithfully than almost anything else in football, because they come from what he believes and how he was taught. When a new coordinator arrives, the most useful question is not "is he good?" but "what did his last offense look like, and who on this roster fits it?"

Before you use that fingerprint, check one thing: **who calls the plays**. A coordinator who doesn't call plays changes much less than one who does. Use the method below in order.

1. **Who calls plays?** The head coach, the coordinator, or someone else? Only the play-caller's fingerprint matters much.
2. **What did his last offense do?** Its neutral PROE, pace, motion and play-action rates, and how it split targets among receivers, tight ends and backs.
3. **What roster did he inherit?** A fingerprint needs players to fit it: a coordinator who feeds the tight end needs a tight end who can run routes.
4. **Who moves up and who moves down?** Translate the changes into shares and volume, position by position.
5. **Check it early, but don't trust it early.** Four games are enough to see a fingerprint, not enough to measure it.

Three 2026 changes make good test cases, because they cover the three kinds of change.

- **Detroit: a new play-caller from a different system.** The Lions are on their third offensive coordinator in three years. Ben Johnson left after 2024 to become the Bears' head coach; his successor, John Morton, lost the play-calling to head coach Dan Campbell midway through 2025 and was let go after the season. The new coordinator is **Drew Petzing**, Arizona's coordinator from 2023 to 2025, and he is expected to call the plays.⁸ Petzing's Arizona offenses threw about a third of their targets to tight ends every year, the most in the league in 2025. Detroit also traded running back David Montgomery to Houston in March 2026, leaving Jahmyr Gibbs as the clear lead back.⁹
- **Los Angeles Chargers: a famous fingerprint arrives.** Jim Harbaugh replaced coordinator Greg Roman with **Mike McDaniel**, fired as Miami's head coach in January 2026 after four seasons. McDaniel calls the plays, and his Miami offenses used pre-snap motion more than almost anyone, on 68-79% of their plays every season from 2022 to 2025.¹⁰

⁸NFL.com, "Lions hiring former Cardinals offensive coordinator Drew Petzing as their new OC": <https://amp.nfl.com/news/lions-hiring-former-cardinals-offensive-coordinator-drew-petzing-as-their-new-oc> ; CBS Sports, "Lions play-caller Drew Petzing fantasy football outlook 2026": <https://fantasy-api.cbssports.com/fantasy/football/news/lions-play-caller-drew-petzing-fantasy-football-outlook-2026/> . Staff history: this book's fact sheet (Morton fired after 2025; Campbell called plays for the second half of 2025). Arizona's tight-end target shares (33% in 2023, 34% in 2024, 35% in 2025): course calculation from nflverse play-by-play. The play-calling role was reported as part of the hire (ESPN, https://www.espn.com/nfl/story/_/id/47662766/sources-lions-finalizing-deal-hire-drew-petzing-new-oc), and September 2026 coverage discusses "Petzing's play-calling" (Fantasy Nerds, Sep. 15, 2026: <https://www.fantasynerds.com/news/story/2026/09/15/detroit-lions-rushing-attack-surprises-in-week-1-victory-over-saints-1623808>); no in-season report says Campbell has taken the calls back, but none states the arrangement outright either.

⁹AP via News4JAX, "Lions trade Montgomery to Texans for Scruggs, 4th, 7th-round picks, AP source says" (Mar. 2, 2026): <https://www.news4jax.com/sports/2026/03/02/lions-trade-montgomery-to-texans-for-scruggs-4th-7th-round-picks-ap-source-says/> . NFL Network reported different compensation; the trade itself is confirmed by nflverse 2026 rosters.

¹⁰Chargers.com, "Mike McDaniel, Chris O'Leary named new coordinators" (2026): <https://www.chargers.com/news/mike-mcdaniel-chris-oleary-new-coordinator-2026> ; AP via News4JAX, "Mike McDaniel joins Harbaugh, Herbert as Chargers' offensive coordinator after Dolphins firing" (Jan. 26, 2026): <https://www.news4jax.com/sports/2026/01/26/mike-mcdaniel-joins-harbaugh-herbert-as-chargers-offensive-coordinator-after-dolphins>

- **Kansas City: a new coordinator, the same play-caller.** Eric Bieniemy returned as the Chiefs' coordinator, replacing Matt Nagy. But Andy Reid has called Kansas City's plays for almost all of his tenure (the exception was the last weeks of 2017, when he handed them to Nagy), including throughout Bieniemy's first stint (2018-2022), and said in January 2026 that play-calling remains his job. The bigger Chiefs change is a player: they signed Kenneth Walker III, the Super Bowl LX MVP, from Seattle in March.¹¹

Here is what the first four weeks of 2026 show.

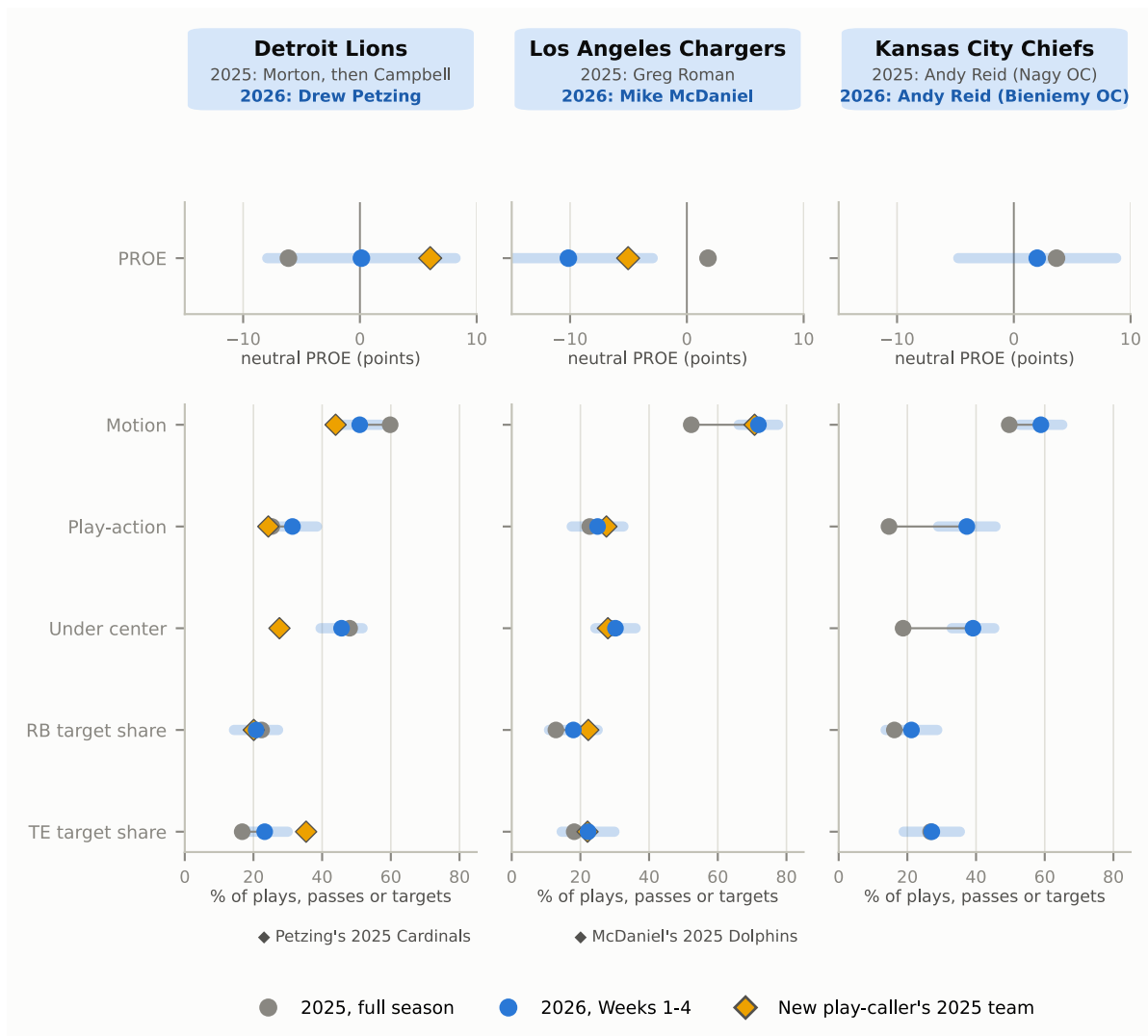


Figure 8: Coordinator-change usage cards: 2025 (grey) against 2026 Weeks 1-4 (blue, with a bar showing about ± 2 standard errors of sampling noise), and the new play-caller's previous team in 2025 (gold diamond) where there is one. Top row: neutral-situation PROE. Bottom row: how the offense plays, as percentages: motion (any motion before or at the snap, FTN definition), play-action (share of passes), under center (share of plays), and the share of targets that went to running backs and to tight ends. Read across a card: where the blue dot has moved toward the gold diamond, the new play-caller's habit has arrived. Four games are a small sample, so a move smaller than the bar is not yet evidence. Data: nflverse play-by-play and FTN charting; regular seasons.

firing/. Miami's motion rates (FTN is [motion](#), motion before or at the snap: 68% in 2022, 79% in 2023, 76% in 2024, 71% in 2025) are this book's calculation from nflverse FTN charting. Roman, the Chargers' OC and play-caller in 2024 and 2025, was fired on Jan. 13, 2026, two days after the wild-card loss to New England: CBS Sports, <https://www.cbssports.com/nfl/news/chargers-fire-offensive-coordinator-greg-roman-five-candidates-to-maximize-justin-herbert/>.

¹¹Chiefs.com, "Chiefs name Eric Bieniemy as offensive coordinator" (Jan. 2026): <https://www.chiefs.com/news/chiefs-name-eric-bieniemy-as-offensive-coordinator>; A to Z Sports, "Andy Reid confirms suspicions about whether new OC Eric Bieniemy will call Kansas City Chiefs' plays in 2026": <https://atozsports.com/nfl/kansas-city-chiefs-news/chiefs-andy-reid-confirms-suspicions-about-whether-new-offensive-coordinator-eric-bieniemy-will-call-plays-in-2026/> (Jan. 26, 2026). Nagy's 2017 stretch: NFL.com, "Andy Reid gives Chiefs play-calling role to Matt Nagy": <https://www.nfl.com/news/andy-reid-gives-chiefs-play-calling-role-to-matt-nagy-0ap3000000886304>. Walker: AP via KPTV, "Chiefs land Super Bowl MVP Kenneth Walker on a 3-year deal worth up to \$45M, AP source says" (Mar. 9, 2026): <https://www.kptv.com/2026/03/09/chiefs-land-super-bowl-mvp-kenneth-walker-3-year-deal-worth-up-45m-ap-source-says/>.

Read the three cards one at a time, the way you would read a scouting report.

Los Angeles: the fingerprint arrived on day one. The Chargers used motion on 52% of their plays in 2025 under Roman. Through four games under McDaniel, the rate is 72%, almost exactly what his last Miami offense did (71%), and the jump is far outside the noise bar. His play-calling has also leaned hard on the run: a neutral PROE of about -10 after $+2$ in 2025. Both are McDaniel habits (his 2025 Dolphins were about -5), but the PROE is noisier than the motion rate, and an offense that starts slowly can run more than it means to. For fantasy, the targets have been spread: no Charger drew even 18% of them through four weeks, and the backs took a bigger share of the targets than in 2025. A spread passing game in a run-first offense is hard on every Chargers receiver's volume. If the run lean holds, the beneficiary is the lead back, Omarion Hampton, who took 52% of the Chargers' runs.

Detroit: a fingerprint plus a roster move. The Lions' tight end target share went from 17% to 23%, toward Petzing's Arizona level, and Sam LaPorta drew about 20% of the targets. Play-action rose from 25% to 31%, and motion fell from 60% to 51%, toward the 44% of Petzing's last Cardinals offense. Each move is about the size of the noise bar, so they are hints rather than proof, but all three point the way the fingerprint predicted. The PROE jump (-6 to about 0) is harder to read, because the Lions trailed by about four points per play on average through four weeks, and trailing pulls even neutral-situation calls toward the pass. The clearest change is not scheme at all: with Montgomery traded, Gibbs took 83% of Detroit's runs, up from 56%. That is a usage change you could have predicted in March.

Kansas City: same play-caller, different ingredients. Reid still calls the plays, so the theory says little should change, and the PROE barely did. But the Chiefs have been under center on 39% of plays (19% in 2025) and have used play-action on 37% of passes (15% in 2025), both far outside the noise. That pattern fits a team built around a new lead back, Walker, who took 76% of the Chiefs' runs, more than fits a new coordinator; a play-caller adjusts to his players as well as the other way round. The lesson is the first step of the method: when the play-caller hasn't changed, look for the roster change instead.

Common misconception

"Four weeks in, the new offense is clearly X." Four games are about 240-260 plays and 120-180 neutral-situation plays per team. With that many plays, a team's PROE has a standard error of about 4 percentage points, so a change of 5 or 6 points can be noise. Rates measured on every play (motion, under center) settle faster than rates measured on a subset (PROE, target shares). Big, consistent moves toward the new play-caller's old numbers (the Chargers' motion rate) are believable early; small moves, or moves against the fingerprint, need another month.

Watch for it on Sunday

Watch for it on Sunday: the scheme-shift draft

Before the season (or now, for the 2026 test cases), pick one team with a new play-caller and write down a prediction for each skill position. Then check it after four weeks and again at mid-season.

Before kickoff (the forecast):

- **Who calls the plays?** If it's the same person as last year, look for roster changes instead.
- **His last offense's fingerprint:** neutral PROE, plays per game, motion and play-action rates, and the share of targets to RBs, TEs and WRs. Which way does each differ from this team's last season?
- **Translate it:** which position's share should rise, which should fall, and does the team's total volume rise or fall? Name the player who gains most and the one who loses most.

During the game (the live check):

- **Who's on the field on third down and in the two-minute drill?** That's the passing-down back and the receivers the play-caller trusts.
- **Does the tight end release or stay in?** Count it for one quarter: on how many dropbacks did he run a route?
- **Who's in at the goal line?** Watch the substitution inside the 5. That player owns the touchdowns.
- **What's the score?** Discount the usage from a blowout: the leading team's carries and the trailing team's targets are game script, not role.

Predict the play

💡 Predict the play 1: first-and-goal at the 1

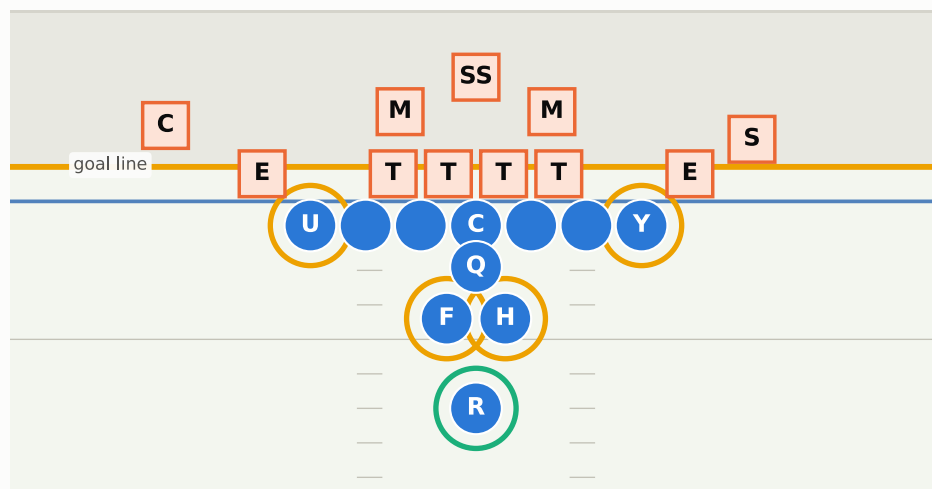


Figure 9: First-and-goal at the 1, Philadelphia, 2024 (illustrative alignment). The offense has loaded up: two tight ends attached to the line (gold rings), the quarterback under center, and two big players (a fullback, F, and a third tight end, H) lined up right behind him, with the running back (R, green ring) a few yards behind them. The defense has packed ten of its eleven players within about three yards of the ball.

Who carries the ball, and what does this play do to the running back's red-zone share and touchdown total?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: third-and-8, down 10 in the fourth

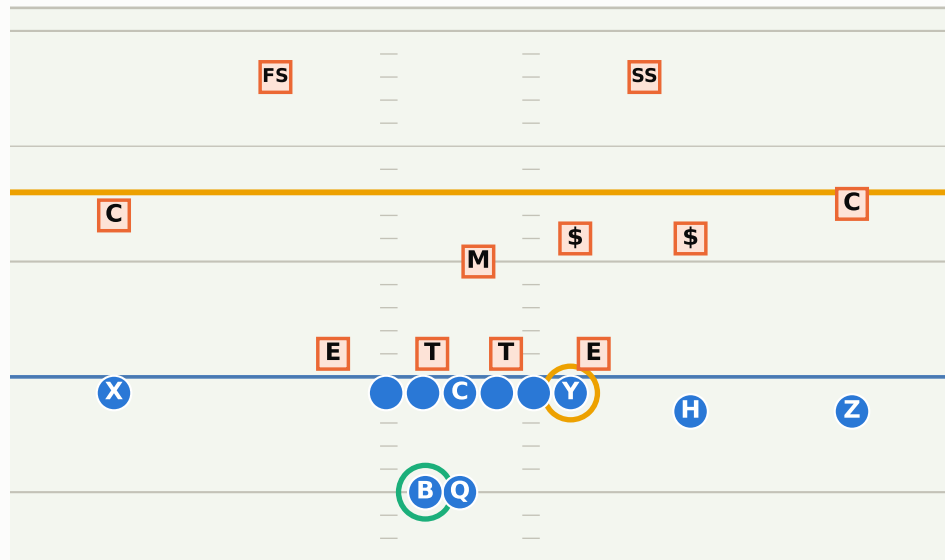


Figure 10: Third-and-8, fourth quarter, the offense trailing by 10. Three receivers to the right (trips, 11 personnel), one to the left, the quarterback in the shotgun and a running back (B, green ring) beside him. This team has two backs: A, its early-down runner, who had 18 carries and 1 catch in the first three quarters, and B. The defense is in dime (six defensive backs), with two safeties deep.

Run or pass? Who on the offense is likely to touch the ball, and what kind of fantasy player is B?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the receiver who changes teams

A wide receiver drew 25% of his team's targets last season, playing for an offense that ran 64 plays a game with a neutral PROE of +5 and averaged 34 targets a game. He signs with a team whose new play-caller's last offense ran 58 plays a game with a PROE of -8 and averaged 27 targets a game. He'll be the clear number-one receiver there too.

Roughly how many targets per game should you forecast for him, and what would make you raise that number during the season?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Fantasy points are outputs; usage is the input the scheme controls.** Opportunity statistics (target share, air-yards share, WOPR, route participation) repeat from season to season far better than efficiency, and touchdowns barely repeat at all.
- **Read receivers in order: on the field, running a route, targeted, targeted deep, productive per route.** Snap share hides blocking snaps; route participation and YPRR fix that, and a productive part-timer whose role grows is the cheapest upside in fantasy.
- **A running back's touchdowns belong to whoever has the goal-line job.** Snap share and carries describe the role between the 20s; red-zone share, and especially who gets the ball inside the 5 and the 2, decides the scoring. In 12 personnel the tight end is a receiver only if he runs routes.
- **Volume = plays × pass rate × share.** PROE sets the team's taste and moves volume, not share; pace adds or subtracts plays for everyone; game script overrides both late in games.
- **Know the format and the price.** PPR rewards catches and lifts pass-catching backs and slot receivers; ADP is a market price, and value is beating it, most often by spotting a usage change the box score hasn't shown yet.

- **When the play-caller changes, his fingerprint travels.** Check who calls plays, what his last offense did, and what roster he inherited, then verify the forecast with a month of data, trusting every-play rates sooner than shares.

Connections

Builds on: Building a Roster (positional value, surplus value and scheme fit, the real-football versions of this chapter's market); Tendencies: PROE, Personnel, Motion, and Coverage (where PROE and the neutral-situation filter come from, and how to measure a team's motion and play-action habits); Working with nflverse Data in Python (the play-by-play, participation and FTN data behind every chart here); Personnel Groupings and Formation Rules and Vocabulary (the tight end's two alignments); Pass Protection (why backs and tight ends stay in); and Red Zone, Goal Line, and Short Yardage (the tush push and the goal-line packages that decide touchdown shares).

Sets up: the Capstone, whose live checklist uses the same on-field reads (who's in on third down, who's in at the goal line, does the tight end release) as part of a full pre-snap routine. The case studies show the fingerprints this chapter measures in their original homes: the Shanahan tree (McDaniel's motion and wide zone), the Chiefs (Reid's play-calling across three coordinators) and the Lions (the offense Petzing inherits).

i Go deeper

Josh Hermsmeyer's air-yards work (archived at FiveThirtyEight and in his later writing) is the foundation of WOPR, RACR and the habit of reading opportunity before production. The nflverse data dictionary documents the `target_share`, `air_yards_share` and `wopr` columns that `load_player_stats()` computes for every player and week, and nflverse's `load_ff_opportunity()` adds expected fantasy points built from usage (each carry and target valued by where it happened), which is this chapter's idea turned into a model. For route-level data, PFF and FTN chart routes run by every player. To extend the chapter's charts, change the season in the data cell, or swap the `neutral()` filter for your own definition and watch how the PROE ranking moves.

Answers to Predict the play

Answer 1.

The quarterback, on the tush push. Everything in the picture says sneak: the quarterback under center with two big players stacked right behind him to push, both tight ends attached, the running back stacked behind the pushers, where a handoff would be slow to reach the line, and a single yard to gain. In 2024 the Eagles ran this from inside the 2 so often that Jalen Hurts got 18 chances there to Saquon Barkley's 6, and scored 11 touchdowns from there to Barkley's 2.

For fantasy, the play means the running back's red-zone share overstates his touchdown chances. Barkley still got the most chances inside the 20 (his carries from the 3 to the 20), but the shortest, surest touchdowns belonged to the quarterback. When you evaluate a back, look up *who gets the ball inside the 2*, not just inside the 20, and check the tush-push rules, which still allow the play in 2026 after the 2025 ban attempt failed.

Answer 2.

A pass, very likely, and B is the passing-down back. The situation alone makes it a pass: third-and-8 is a passing down at any score, and in 2025 teams trailing by 9 to 15 points in the fourth quarter threw on about four plays in five even before you add the down and distance. The personnel agree: three wide receivers, the quarterback in the gun, and the defense in dime, expecting a throw.

The back on the field tells you about usage. The team swapped its early-down runner, A, for B, a back it trusts to pick up a blitz and run a route, which is why B is on the field when the team must throw. Expect B to stay in to block if the defense shows pressure, and to release to the flat or over the middle as a checkdown if it doesn't; the slot receivers (H and Y on the trips side) are the other likely targets on a third-down throw short of the sticks. For fantasy, B is the classic PPR back: few carries, but targets in exactly the game scripts where his team trails. In standard scoring he is barely rosterable; in PPR, on a team that trails often, he can be a weekly starter, the 2025 Kenny Gainwell profile. A, meanwhile, is a game-script player: valuable when his team leads, invisible when it doesn't.

Answer 3.

About 7 targets a game, down from about 8.5. Start with the part that travels: his role. A clear number-one receiver keeps a share in the mid-20s on almost any team (the left panel of the PROE chart: share doesn't depend on the team's taste), so keep 25%.

Then apply the new team's volume: 25% of 27 targets is about 6.8 a game, against 25% of 34, about 8.5, before. That's a drop of about a fifth, roughly 25-30 targets over a 17-game season, with no change in his talent at all. In PPR, at a typical two-thirds catch rate, that's 16-20 fewer catches before you count a single lost yard.

Raise the forecast if any term of the volume equation moves his way in the first month: the new team throwing more than its play-caller's history (a higher PROE, possibly because it is trailing, which is game script, not taste, and less reliable); a faster pace; or a target share above 28-30%, meaning the play-caller is building the passing game around him more than his last team did. Lower it if a second receiver or the tight end takes a bigger share than expected.